

datum

build guide

made by Christian Faurholt

Before you start

Critical notes

■ The PCB is reversible.

The same PCB is used for both left and right halves. Jumper pads determine which side it becomes. You must close the correct jumper pads before soldering anything else.

■ Battery polarity is critical.

Reversed polarity will instantly and permanently damage the Nice!nano. Always verify with a multimeter before soldering.

■ Never solder the Nice!nano directly.

Always use Mill-Max sockets. This allows you to remove the controller for firmware flashing or replacement.

Jumper pads

The datum PCB has jumper pads that configure it as either a left or right half.

- Left half — close the jumper pads on the top side of the PCB
- Right half — close the jumper pads on the bottom side of the PCB

Bridge the pads with a small amount of solder. Verify with a multimeter in continuity mode before proceeding.

Soldering order

Always solder in this order. Components soldered out of order can be impossible to rework later.

Step 1 — Power switches

Place the Alps SPDT power switch matching the footprint outline and solder all three legs. Orientation matters — match the silkscreen. Verify with a multimeter: in the OFF position there should be no continuity between BAT+ and RAW.

Step 2 — Reset buttons

Place the Panasonic SMD reset button matching the footprint and solder all four legs. No orientation required — it is symmetric.

Step 3 — Mill-Max sockets

Insert sockets from the top side. Place tape over the openings to prevent solder wicking into the pins. Solder from the underside. Remove tape after soldering. Verify each socket is flush. Do not insert the

Nice!nano yet.

Step 4 — Battery

Verify polarity with a multimeter before soldering. Red wire = BAT+ (positive). Black wire = BAT- (negative/ground). Measure wire length before cutting — battery sits flat under the Nice!nano. Solder red to BAT+ and black to BAT-. Double-check polarity before proceeding.

Step 5 — Switches

Solder all 17 switches per half. Press each switch firmly until it clicks, then solder both pins. Before soldering, verify each position by shorting the pads with tweezers.

Step 6 — Nice!nano

Insert into Mill-Max sockets. Components face down toward the PCB. USB-C port faces toward the inner edge of the board. Do not solder — it sits in the sockets and can be removed.

Tenting

Start at 10–15 degrees. Lock the Manfrotto angle firmly — SKUF feet prevent sliding.

Troubleshooting

A key does not register

Check switch pins are fully soldered. Short the switch pads with tweezers — if the key registers, the switch is faulty.

An entire row or column does not register

Check solder joints on the controller pads in that row or column.

The keyboard does not power on

Check power switch orientation and verify battery polarity.

The halves do not connect wirelessly

Both halves must be powered on. Left half is central. If pairing fails, clear ZMK bonds and re-pair.

Nice!nano not detected when flashing

Double-tap reset quickly. Try a different USB-C cable — some cables are charge-only.